

Scriptje

In our research we have looked at the crab nebula. This nebula has been around since 1054, when a massive star went supernova. A star stays stable through gravity and outward pressure of fusion in the core. When a massive star has exhausted its nuclear fuel, the outward pressure falls away. The star cannot balance the gravity anymore and the core collapses. This causes a shockwave which expels the material into space at high velocities, creating a supernova.

Ster rondje met pijltjes gravity naar binnen en pressure naar buiten en dan buiten pressure pijltje weg en ster rontje word kleiner tot die bijna weg is en dan explodeert die in een andere kleur en wordt een heel groot rondje

We've observed the remnants of the supernova crab nebula. We've used the telescopes at the Anton Panenkov Institute. There are different types of filters to be used. You have the broadband filters and narrowband filters. We used the narrowband filters H α , SII and OIII.

Telescoop foto. Spectrum met narrowband en pijltje die ernaar wijst en H α enzo erbij zet

These are the different narrowband images of SII, H α , and OIII. To see where each of the elements are situated you can combine these images into one image. However to get a better understanding of the distribution and processes of the nebula, we combined two of the narrowband filters and looked at the intensity ratio of each of the combinations. We've compared this with previous study's to find the explanations for the intensity ratio's. Let's first look at SII and H α . EXPLANATION BLA BLA BLA

Foto's op een rijtje en gaan over elkaar heen en worden een nieuwe combinatie

Now let's look at the second combination OIII/SII. EXPLANATION BLA BLA BLA

And now the last combination OIII/H α . Explanation...